COOKING WITHOUT POWER

Feeding the household safely during an outage: what to eat first, how to cook outdoors, and how to keep food safe.

DOCUMENT TR-06
SERIES TRAINING
REVISION B // NOV 2026
STATUS PUBLIC ISSUE

GUIDE / VERIFY: General preparedness information, not a substitute for professional training, local rules, manufacturer instructions or medical care. **Practice skills in safe conditions before you need them, and verify important information with current official sources.**

COOK OUTDOORS. Grills, camp stoves and other fuel-burning cookers make carbon monoxide. Never use them inside a home, garage, basement, tent or camper, even with doors and windows open, unless the appliance's instructions specifically say indoor use is allowed. Never bring a grill indoors to heat the house.

1. THE ORDER TO EAT IN

#	FOOD	WHY
1	Refrigerator	Safe for about **4 hours** without power if the door stays closed. Eat perishables first.
2	Freezer	A full freezer keeps food frozen about **48 hours** (24 if half full). Food that still has ice crystals, or is at 40°F or below, can be refrozen or cooked.
3	Pantry and stored food	Lasts the longest, so use it last. *(TL-02)*

Throw out perishable food (meat, poultry, fish, eggs, milk, leftovers) that has been above **40°F for 2 hours or more**. Never taste food to find out. **When in doubt, throw it out.**

2. WAYS TO COOK

METHOD	USE IT	SAFETY
Gas grill (propane)	Pots and pans work on most grills. Good for boiling water, too.	Outdoors, well away from the house, deck rails and overhanging branches. Check hoses for leaks with soapy water.
Charcoal grill	Slow to start, but holds heat well. Cook with the lid on to save fuel.	Outdoors only. Let coals cool completely, then put them in a metal can. Never use charcoal indoors: it gives off deadly CO.
Camp stove	Fast, uses small fuel canisters. The best all-around backup cooker.	Outdoors unless the maker says otherwise. Keep a spare fuel canister. Stable, level surface.
Fire pit or fireplace	Cast-iron pots and foil packets over coals.	Follow fire rules. *(TR-01)* Only cook in a fireplace with a working chimney.
Solar oven	Slow cooking on bright, sunny days. Uses no fuel.	Use a food thermometer: cooking takes hours, and food must get hot enough to be safe.
No cooking	Canned food, sandwiches, nut butter, cereal with shelf-stable milk.	Keep a few no-cook days of food for when cooking isn't safe.

3. HOW LONG WILL MY FUEL LAST?

Hours of cooking = energy in the fuel ÷ burner's rating (BTU per hour, printed on the stove). Real-world use is usually shorter: wind, cold and simmering change it. Test your stove: time how long one canister lasts.

FUEL	ENERGY (APPROX.)	ON A 10,000 BTU/HR BURNER, ON HIGH
1 lb propane cylinder (camp size)	about 21,500 BTU	about 2 hours
20 lb propane tank (grill size)	about 430,000 BTU	about 43 hours
My stove: _____ BTU/hr x _____ hours a day		Fuel needed for ____ days:

4. COOK IT SAFELY

Use a food thermometer. These are the safe minimum internal temperatures:

FOOD	COOK TO
Poultry (whole or ground)	165°F
Ground beef, pork, lamb	160°F
Beef, pork, lamb steaks, chops, roasts	145°F, then rest 3 min
Fish	145°F
Egg dishes	160°F
Leftovers and casseroles	165°F

- Wash hands with soap and safe water before cooking. If water is short, use hand sanitizer, then wash when you can.
- Keep raw meat away from ready-to-eat food.
- Cook only what you'll eat; leftovers are hard to keep cold.
- Save water: one-pot meals, foil packets, paper plates.
- Use only safe water for cooking and drinks. *(TR-02)*
- Keep a fire extinguisher nearby. Keep children and pets away from the cooking area.

5. STORING FUEL

- Store propane cylinders and tanks **outdoors**, upright, in a shaded, ventilated spot. Never in the house, basement or garage.
- Keep charcoal dry, in a closed metal container. Store lighter fluid away from heat, and never add it to a fire that's already lit.
- Buy only the fuel your stove is made for. Check expiration dates on fuel canisters and the condition of hoses each season.

SOURCES / FURTHER REFERENCE

1. USDA Food Safety and Inspection Service / FoodSafety.gov: Food Safety in a Power Outage; Safe Minimum Internal Temperatures. `foodsafety.gov`
2. Centers for Disease Control and Prevention: Avoiding Carbon Monoxide Poisoning; Keep Food Safe After a Disaster. `cdc.gov`
3. National Fire Protection Association: Grilling Safety. `nfpa.org`
4. FEMA: Ready.gov: Power Outages; Food. `ready.gov`
5. U.S. Energy Information Administration: energy content of propane (about 91,500 BTU per gallon). `eia.gov`

IMPORTANT: Follow the manufacturer's instructions for every stove, grill and fuel. This document reflects the sources above as of November 2026.